

Practice Registering Medical Volumes

The example you just saw is provided for you in the course files. Navigate to the Module 2 folder and open the **RegisteringMedicalVolumes.mlx** file to try the example yourself.

We recommend working with the code in MATLAB to experiment with the code and explore the results in more detail. A static version of the code is available below. The code will not look the same here as interactive tools are used in MATLAB.

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Visualize the resampled volume

Visualize a slice of the volumes you will register. The volumes are not aligned, but the resampling has brought them much closer together. You can see that there is both rotation and translation between the two images. Less obvious is that there is also scale, which is usually the case the images are acquired at different resolutions. This means you'll need a similarity transformation.

Because the resampled 3 Tesla volume is better positioned, we'll use it as the fixed volume and the 7 Tesla volume as the moving volume for registration.

```
img1 = extractSlice(resampled3T, 84, "transverse");  
img2 = extractSlice(vol7T, 84, "transverse");  
imshowpair(img1, img2, "montage")
```

Register the Volumes

To use the `imregister` function, you must create an optimizer and a metric with the `imregistercfg` function. These objects include properties you can adjust if necessary to improve the registration. The default settings are a good starting point and often work well. Use the `multimodal` option because the images were acquired with very different machines.

The code below generates a new 3D array, `newVoxels` that is 7 Tesla volume registered to the resampled 3 Tesla volume.

```
[optimizer,metric] = imregistercfg("multimodal");  
newVoxels = imregister(vol7T.Voxels, resampled3T.Voxels, "similarity", optimizer,  
metric);
```

Create a new volume with the registered data

The `imregister` does not create a `medicalvolume` object, but the metadata associate with the volume is important. Create a new `medicalvolume` that is a copy of the original. Then, replace the `voxels` property with the new registered voxels.

```
registered7T = vol7T;  
registered7T.Voxels = newVoxels;
```

View results

Visualization is an important part of evaluating your results. Use the controls below to compare slices in different planes. The volumes are well aligned!

```
slicenum = 84;  
plane = "transverse";  
imgresampled3T = extractSlice(resampled3T, slicenum, plane);  
img7Treg = extractSlice(registered7T, slicenum, plane);  
imshowpair(imgresampled3T, img7Treg)
```

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